

Scalability & Future Plan

Date	08 October 2026
Team ID	SWTID-2026-2386
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	1 Mark

Current System Limitations

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	No saved history or user accounts	Students lose their answers when they close the page	Medium
2	Free-tier Gemini limits and AI latency	Slow or rate-limited replies when many students use it at once	High
3	Text only; no voice, files or images	Students cannot upload a PDF or a photo of a problem	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single worker; stateless, so more workers can be added	Several Uvicorn/Gunicorn workers behind Nginx; container autoscaling
2	Data Storage	None	PostgreSQL for accounts, history and quiz scores
3	Performance	One Gemini call per request	Cache repeated questions, use the async Gemini client, rate-limit each user
4	Security	Key in .env, input limits, safe error messages	Secrets manager, HTTPS-only deployment, per-user rate limiting

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Voice input and output, multilingual support	1-2 months	More accessible to learners in different languages
Phase 3	Progress tracking dashboard, badges, learning streaks, adaptive learning paths	2-3 months	More engagement and personalised practice
Phase 4	PDF and image input, mobile app, teacher/parent dashboards, LMS integration (Moodle, Google Classroom)	3-6 months	Wider reach inside schools and colleges